

Inhibition of non-haem iron absorption in man by polyphenolic-containing beverages.



The effects of different polyphenol-containing beverages on Fe absorption from a bread meal were estimated in adult human subjects from the erythrocyte incorporation of radio-Fe. The test beverages contained different polyphenol structures and were rich in either phenolic acids (chlorogenic acid in coffee), monomeric flavonoids (herb teas, camomile (*Matricaria recutita* L.), vervain (*Verbena officinalis* L.), lime flower (*Tilia cordata* Mill.), pennyroyal (*Mentha pulegium* L.) and peppermint (*Mentha piperita* L.), or complex polyphenol polymerization products (black tea and cocoa). All beverages were potent inhibitors of Fe absorption and reduced absorption in a dose-dependent fashion depending on the content of total polyphenols. Compared with a water control meal, beverages containing 20-50 mg total polyphenols/serving reduced Fe absorption from the bread meal by 50-70%, whereas beverages containing 100-400 mg total polyphenols/serving reduced Fe absorption by 60-90%. Inhibition by black tea was 79-94%, peppermint tea 84%, pennyroyal 73%, cocoa 71%, vervain 59%, lime flower 52% and camomile 47%. At an identical concentration of total polyphenols, black tea was more inhibitory than cocoa, and more inhibitory than herb teas camomile, vervain, lime flower and pennyroyal, but was of equal inhibition to peppermint tea. Adding milk to coffee and tea had little or no influence on their inhibitory nature. Our findings demonstrate that herb teas, as well as black tea, coffee and coca can be potent inhibitors of Fe absorption. This property should be considered when giving dietary advice in relation to Fe nutrition.